Forensic Analysis of UAV Artifacts Using the NIST CFReDS Dataset

Field	Details
Student	Stella Hyatt
Course Project	CS 426 Project
Project Type	Digital forensics investigation / dataset analysis
Primary Evidence Source	NIST CFReDS DroneDataSet, Steve Watson / VTO Inc.
Tools Used	FTK Imager, CsvView, file system review, screenshots, and project documentation
Report Date	April 25, 2026

Purpose Statement

This report documents a basic but complete digital forensic workflow performed on a drone dataset. The goal of the project was not to build a software application. Instead, the project focused on examining provided drone evidence, identifying accessible artifacts, reviewing flight log data, and explaining how forensic tools can help recover and interpret information from drone storage media.

1. Executive Summary

This project investigated drone forensic evidence from the NIST Computer Forensic Reference Data Sets (CFReDS) DroneDataSet. The dataset was created from a controlled drone forensics program in which drones and related devices were operated, acquired, imaged, and examined for artifacts. The project used a practical forensic workflow to inspect the dataset, document the file structure, and analyze available UAV flight log files.

The main evidence reviewed for this project consisted of DAT flight log files located in the dataset flight_logs folder. FTK Imager was used to view the evidence source and demonstrate the file-level forensic approach, while CsvView was used to parse and interpret the DAT flight logs as UAV log data. CsvView provided the most meaningful analysis because it converted raw log data into readable categories and values that could be discussed as flight evidence.

The project demonstrates that drone evidence can contain information beyond ordinary files. Even when a forensic imaging tool mainly displays low-level hexadecimal or file metadata, specialized log viewers can reveal operational details such as recorded flight activity, telemetry-style fields, sensor-related data, and structured log categories. The final deliverables include screenshots, notes, a GitHub repository with organized materials, and this written report.

2. Background and Dataset Description

The evidence source for this project was the NIST CFReDS DroneDataSet associated with Steve Watson / VTO Inc. According to the dataset description used for the project, the Drone Forensics Program involved purchasing and operating multiple drone models in a controlled, geofenced environment. After operation, attempts were made to acquire and image storage areas from the drones, controllers, connected mobile devices, and computers. Some drones were also disassembled so investigators could identify internal storage areas.

This background is important because the project is not based on a simulated app or a program created by the student. It is based on real forensic-style reference data. The role of the investigator is to locate useful artifacts, preserve what

is found, and explain what the artifacts show. In this case, the focus was on drone flight logs and how those logs can be interpreted using forensic and analysis tools.

3. Project Objective

- Identify the relevant evidence files within the drone dataset.
- Use FTK Imager to demonstrate forensic file review and low-level evidence inspection.
- Use CsvView to open and interpret DAT flight log files as UAV logs.
- Document findings with screenshots, notes, and a written explanation of the forensic workflow.
- Prepare a clear final presentation explaining that the project is a forensic investigation, not an application development project.

4. Tools and Their Purpose

Tool	Purpose in the Project
FTK Imager	Used to open and inspect the evidence folder/file structure. It helped demonstrate a forensic review process and showed how raw evidence may appear at the file and hexadecimal level.
CsvView	Used to open DAT flight logs. It recognized the logs as UAV data and provided readable sections, values, and categories for analysis.
GitHub	Used to organize final deliverables, screenshots, notes, and README documentation for the project.
Screenshots / Notes	Used to preserve visible evidence from the analysis process and support the final presentation.

5. Evidence Overview

The primary evidence reviewed was a collection of DAT files located in the flight_logs folder. These files were too large to upload directly to GitHub, so the repository was organized around supporting materials instead of storing every raw log file. The deliverables folder includes screenshots and documentation showing the analysis process.

- Evidence type: Drone flight log files in DAT format.
- Example analyzed file: FLY015.DAT.
- Observed behavior: CsvView successfully opened the file and recognized it as a UAV log.
- Repository decision: Large DAT files were excluded from GitHub, while screenshots and notes were preserved as deliverables.

6. Methodology

The forensic process followed a simple repeatable workflow. The goal was to preserve the original dataset, inspect the available artifacts, and document what could be meaningfully interpreted.

1. Downloaded and organized the drone dataset from NIST CFReDS.
2. Located the flight_logs folder containing DAT flight log files.
3. Opened the evidence folder in FTK Imager to review file structure and demonstrate forensic inspection.
4. Selected a representative DAT file, such as FLY015.DAT, for closer review.
5. Attempted to view raw contents in FTK Imager, noting that the tool mainly displayed low-level data rather than interpreted flight information.

6. Opened the DAT file in CsvView, where the file was recognized as a UAV log.
7. Reviewed the available CsvView sections and captured screenshots of important categories and values.
8. Organized screenshots, README files, notes, and report materials in the project repository.

7. FTK Imager Analysis

FTK Imager was useful for showing the forensic side of the project. It allowed the evidence folder and selected files to be viewed without treating the files like normal user documents. In this project, FTK Imager mainly showed that the DAT files existed, could be selected, and contained raw data. For the DAT files, FTK Imager did not provide a detailed interpretation of the flight log contents. Instead, it displayed low-level information such as file bytes and hexadecimal-style data.

Overall, this limitation is still valuable in a forensic report. It shows that not every forensic tool automatically explains every artifact. FTK Imager helped establish the presence of evidence and supported a file-level walkthrough, while CsvView provided the specialized interpretation needed for the flight logs.

8. CsvView Flight Log Analysis

CsvView was the main analysis tool for the project because it was able to interpret the DAT files as UAV flight logs. When FLY015.DAT was opened, CsvView recognized the file as a UAV log and displayed structured data instead of only raw bytes. This made it possible to explain the flight log as evidence rather than just as an unknown binary file.

In the presentation, the CsvView portion should be treated as the strongest evidence demonstration. The important point is that CsvView turns the drone log into readable categories, allowing the investigator to describe what kinds of information a drone may record during operation.

8.1 CsvView Sections to Discuss

CsvView Area	How It Supports the Investigation
Log Recognition / Summary Area	Shows that the selected DAT file is recognized as a UAV/UAS log. This supports the claim that the file is relevant drone evidence.
Message or Data Categories	Displays grouped log categories. These categories help organize the flight data and show that the file contains structured operational records.
Telemetry-Style Values	Can include changing values associated with movement, positioning, altitude, speed, orientation, or sensor states, depending on what the log contains.
Graphs / Plots	Useful for visually showing that the log contains changing flight-related values over time.
CSV/Table View	Useful for explaining that the log can be reviewed as rows and columns of data, which is easier to document than raw hexadecimal output.

9. Findings

Finding	Explanation
Finding 1	The dataset contained drone flight log files in DAT format, which were treated as the primary evidence artifacts for the project.
Finding 2	FTK Imager was able to support a forensic walkthrough

	of the evidence location and selected files, but it did not convert the DAT file into readable flight information.
Finding 3	CsvView successfully recognized FLY015.DAT as a UAV log, making it the most useful tool for interpreting the flight data.
Finding 4	The analysis showed that drone data can contain structured operational records that are not obvious when viewed only as raw files.
Finding 5	The project produced documentation and screenshots that explain the forensic workflow and provide evidence for the final presentation.

10. Forensic Significance

Drone forensics is important because drones may store information about operation, movement, connected devices, and user activity. Even if an investigator cannot immediately understand a raw file, the artifact may still contain valuable evidence. This project demonstrates the importance of using both general forensic tools and specialized viewers.

The DAT flight logs matter because they represent recorded activity from the drone. In a real investigation, this type of data could help answer questions about whether a drone was operated, when it may have been used, what type of information was recorded, and whether useful artifacts remained after acquisition or teardown. The project does not claim to prove a real crime or incident. Instead, it demonstrates how reference data can be examined using a forensic workflow.

11. Limitations

- The project focused on a limited set of flight logs rather than every artifact in the full dataset.
- The raw DAT files were too large to include directly in the GitHub repository.
- FTK Imager did not automatically interpret the flight log contents, so a specialized tool was required.
- The report avoids claiming exact flight paths, exact coordinates, or exact aircraft behavior unless those values are directly visible in the student screenshots or CsvView output.
- The project is a forensic analysis and documentation project, not a custom-built software application.

14. Conclusion

This project successfully demonstrates a basic drone forensics workflow using the NIST CFReDS DroneDataSet. The investigation identified DAT flight logs as the primary artifacts, used FTK Imager for forensic evidence inspection, and used CsvView to interpret the logs as UAV data. The strongest result was that CsvView could recognize and display FLY015.DAT as a UAV log, making the raw drone evidence understandable and presentable.

Overall, the project shows that drone forensic analysis often requires more than simply opening files. A general forensic tool can preserve and inspect evidence, but specialized viewers may be necessary to interpret flight records. The final project deliverables document the workflow, explain the significance of the findings, and provide a clear basis for a six-minute presentation and question-and-answer discussion.

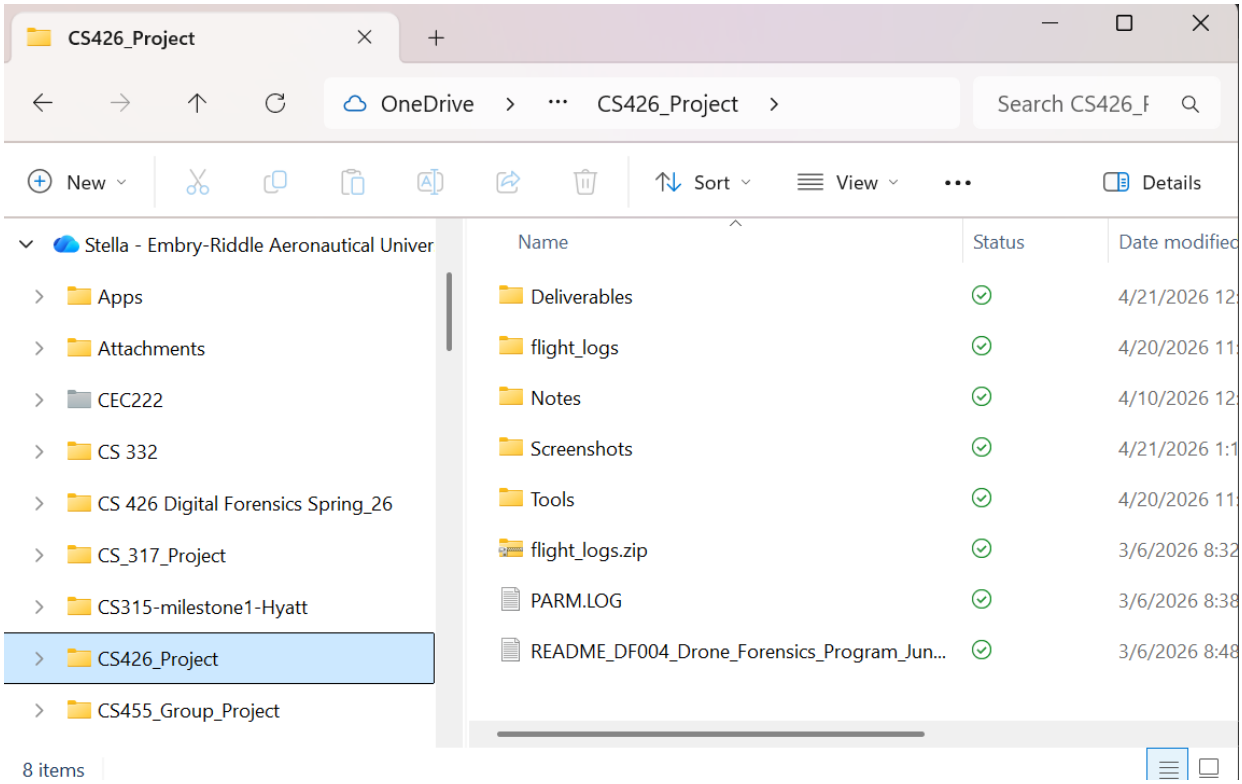
15. References

9. National Institute of Standards and Technology. Computer Forensic Reference Data Sets (CFReDS) Portal. https://cfreds.nist.gov/

10. National Institute of Standards and Technology. CFReDS DroneDataSet, Steve Watson / VTO Inc. https://cfreds.nist.gov/all/SteveWatson%2FVTOInc./Dronedataset
11. FTK Imager. Exterro / AccessData forensic imaging and evidence preview tool.
12. CsvView. UAV log viewing and analysis tool used to interpret DAT flight log files.

Appendix A: Suggested Screenshot Checklist

- Screenshot of the dataset folder structure.



- Screenshot of the flight_logs folder containing DAT files.

OneDrive > Stella - Embry-Riddle Aeronautical University > CS426_Project > flight_logs

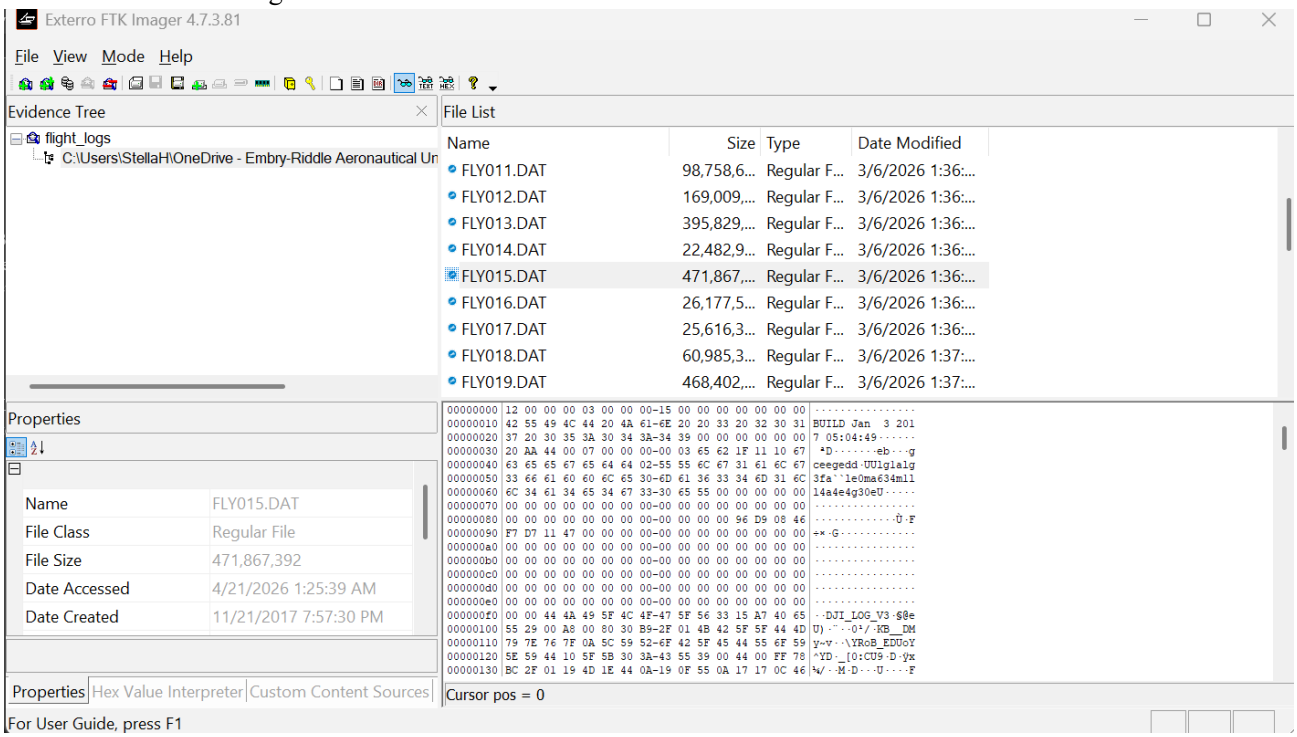
Search flight_logs

Sort View

Name	Status	Date modified	Type	Size
FLY003.DAT		3/6/2026 8:35 AM	DAT	227,084 KB
FLY004.DAT		3/6/2026 8:35 AM	DAT	299,724 KB
FLY005.DAT		3/6/2026 8:35 AM	DAT	402,416 KB
FLY006.DAT		3/6/2026 8:35 AM	DAT	142,940 KB
FLY007.DAT		3/6/2026 8:35 AM	DAT	180,480 KB
FLY008.DAT		3/6/2026 8:36 AM	DAT	376,836 KB
FLY009.DAT		3/6/2026 8:36 AM	DAT	124,332 KB
FLY010.DAT		3/6/2026 8:36 AM	DAT	217,336 KB
FLY011.DAT		3/6/2026 8:36 AM	DAT	96,444 KB
FLY012.DAT		3/6/2026 8:36 AM	DAT	165,048 KB
FLY013.DAT		3/6/2026 8:36 AM	DAT	386,552 KB
FLY014.DAT		3/6/2026 8:36 AM	DAT	21,956 KB
FLY015.DAT		3/6/2026 8:36 AM	DAT	460,808 KB
FLY016.DAT		3/6/2026 8:36 AM	DAT	25,564 KB
FLY017.DAT		3/6/2026 8:36 AM	DAT	25,016 KB
FLY018.DAT		3/6/2026 8:37 AM	DAT	59,556 KB
FLY019.DAT		3/6/2026 8:37 AM	DAT	457,424 KB
FLY020.DAT		3/6/2026 8:37 AM	DAT	77,116 KB
FLY021.DAT		3/6/2026 8:37 AM	DAT	302,540 KB
FLY022.DAT		3/6/2026 8:37 AM	DAT	57,352 KB
FLY023.DAT		3/6/2026 8:37 AM	DAT	166,092 KB
FLY024.DAT		3/6/2026 8:37 AM	DAT	140,564 KB
FLY025.DAT		3/6/2026 8:37 AM	DAT	82,040 KB
FLY026.DAT		3/6/2026 8:37 AM	DAT	195,464 KB
FLY027.DAT		3/6/2026 8:37 AM	DAT	179,596 KB
FLY028.DAT		3/6/2026 8:38 AM	DAT	212,144 KB
FLY029.DAT		3/6/2026 8:38 AM	DAT	29,148 KB
FLY030.DAT		3/6/2026 8:38 AM	DAT	27,712 KB
FLY031.DAT		3/6/2026 8:38 AM	DAT	64 KB
FLY032.DAT		3/6/2026 8:38 AM	DAT	64 KB
FLY033.DAT		3/6/2026 8:38 AM	DAT	60,540 KB
FLY034.DAT		3/6/2026 8:38 AM	DAT	39,284 KB
FLY035.DAT		3/6/2026 8:38 AM	DAT	93,584 KB
FLY036.DAT		3/6/2026 8:38 AM	DAT	419,928 KB
FLY037.DAT		3/6/2026 8:38 AM	DAT	11,516 KB
FLY038.DAT		3/6/2026 8:38 AM	DAT	235,764 KB

FLY039.DAT	✓	3/6/2026 8:38 AM	DAT	370,136 KB
FLY040.DAT	✓	3/6/2026 8:38 AM	DAT	8,600 KB

- Screenshot of FTK Imager with the selected evidence folder or DAT file visible.



- Screenshot of CsvView graph or table-style output, if available.

eventLog::FLY015

```
1142.948 :   63304 [L-ESC][motor_on_comm]sender=0x02 index=0x00 set=0x00 id=0x01
1142.948 :   63304 [L-ESC][motor_on_comm]sender=0x02 index=0x00 set=0x00 id=0x01
1143.947 :   63354 [L-ESC][motor_on_comm]sender=0x02 index=0x00 set=0x00 id=0x01
1143.967 :   63355 [L-ESC][motor_on_comm]sender=0x02 index=0x00 set=0x00 id=0x01
1143.967 :   63355 [L-ESC][motor_on_comm]sender=0x02 index=0x00 set=0x00 id=0x01
1143.968 :   63355 [L-ESC][motor_on_comm]sender=0x02 index=0x00 set=0x00 id=0x01
1144.451 :   63380 [L-FMU/FSM]state changed. cur: in air
1144.453 :   63380 [L-FDI][FDI CTRL] event: impact turn on
1144.470 :   63381 [L-HOME]Update. lati:39.9858755 longti:-105.0077672 alti:1575.67
1144.472 :   63381 [L-FLYMODE]CTRL reset all by assisted takeoff finish
1144.491 :   63382 [L-FLYMODE][Ctrl<1>] REQ_RC_NORMAL ATTI_HOLD ctrl_gps_atti
1144.967 :   63405 [L-ESC][motor_on_comm]sender=0x02 index=0x00 set=0x00 id=0x01
1144.987 :   63406 [L-ESC][motor_on_comm]sender=0x02 index=0x00 set=0x00 id=0x01
1144.987 :   63406 [L-ESC][motor_on_comm]sender=0x02 index=0x00 set=0x00 id=0x01
1145.007 :   63407 [L-ESC][motor_on_comm]sender=0x02 index=0x00 set=0x00 id=0x01
1145.272 :   63421 [L-FMU/LED]action changed. set home:(0)
1145.546 :   63434 [L-FMU/FSM]not near ground
1145.987 :   63456 [L-ESC][motor_on_comm]sender=0x02 index=0x00 set=0x00 id=0x01
1146.007 :   63457 [L-ESC][motor_on_comm]sender=0x02 index=0x00 set=0x00 id=0x01
1146.007 :   63457 [L-ESC][motor_on_comm]sender=0x02 index=0x00 set=0x00 id=0x01
1146.027 :   63458 [L-ESC][motor_on_comm]sender=0x02 index=0x00 set=0x00 id=0x01
1147.239 :   63519 [L-NS][AHRS] yaw aligned 1
1147.372 :   63526 [L-FMU/LED]action changed. Normal Flash(0)
1147.372 :   63526 [L-FMU/LED]type:0, normal flash action:
1147.372 :   63526 [L-FMU/LED]c0:0,15;c1:0,3;c2:0,13;c3:2,3;c4:0,10;c5:0,3;c6:0,10;c7:0,3
1150.551 :   63685 [L-FMU/MOTOR]safe_near_grd:1
1165.372 :   64426 [L-FMU/LED]action changed. battery warning!:battery_status_data.capacity_level1_s
1232.811 :   67798 [L-RC]Emergency break. rc vib!
```

Be synced by other Players | Synch other Players | Synch and be synced other Players | Don't synch with other Players

GeoPlayer::FLY015

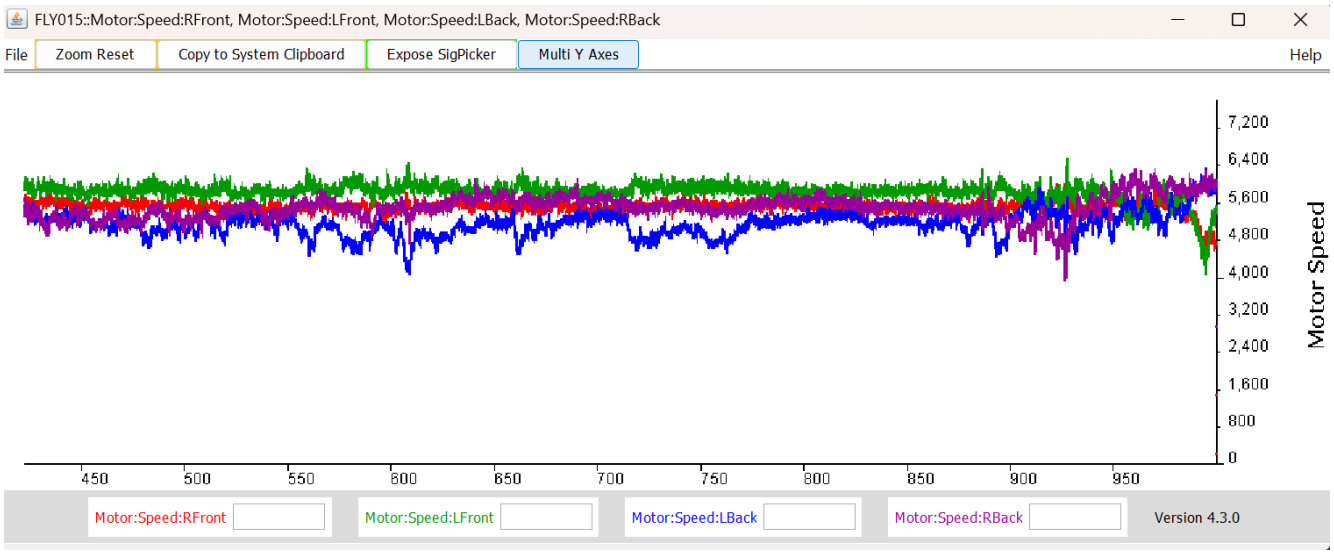
Zoom Reset | GoogleMap Key | Copy to System Clipboard

IMU0 Choose Color | IMU_ATTI(0):yaw:C Choose Color

IMU_ATTI(0):magYaw:C Choose Color | IMU_ATTI(0):tiltDirectionEarthFrame:C Choose Color

AirSpeed:windDirection Choose Color

IMUCalcs(0) Choose Color | IMU_ATTI(0):yaw:C Choose Color


FLY015::Motor:Speed:RFront, Motor:Speed:LFront, Motor:Speed:LBack, Motor:Speed:RBack
File
Zoom Reset
Copy to System Clipboard
Expose SigPicker
Multi Y Axes
Help
7,200
6,400
5,600
4,800
4,000
3,200
2,400
1,600
800
0
Motor Speed
450
500
550
600
650
700
750
800
850
900
950
Motor:Speed:RFront
Motor:Speed:LFront
Motor:Speed:LBack
Motor:Speed:RBack
Version 4.3.0